The Physics Teacher

PAPERS | DECEMBER 01 2017

Pendulum Underwater – An Approach for Quantifying Viscosity

José Costa Leme; Agostinho Oliveira

Check for updates

Phys. Teach. 55, 555–557 (2017)
https://doi.org/10.1119/1.5011833

View Online
Export Citation

Articles You May Be Interested In

Dynamics and performance of clock pendulums

Am. J. Phys. (November 2014)

Quantitative analysis of a smartphone pendulum beyond linear approximation: A lockdown practical homework

Am. J. Phys. (May 2022)

Interplay between Airy and Coriolis precessions in a real Foucault pendulum

Am. J. Phys. (April 2025)

01 April 2026 11:37:20

Pendulum Underwater – An Approach for Quantifying Viscosity

José Costa Leme, EB 23S de Lanheses, Portugal
Agostinho Oliveira, EB/S Sidónio Pais, Caminha, Portugal

The purpose of the experiment presented in this paper is to quantify the viscosity of a liquid. Viscous effects are important in the flow of fluids in pipes, in the bloodstream, in the lubrication of engine parts, and in many other situations. In the present paper, the authors explore the oscillations of a physical pendulum in the form of a long and lightweight wire that carries a ball at its lower end, which is totally immersed in water, so as to determine the water viscosity. The system used represents a viscous damped pendulum and we tried different theoretical models to describe it. The experimental part of the present paper is based on a very simple and low-cost image capturing apparatus that can easily be replicated in a physics classroom. Data on the pendulum's amplitude as a function of time were acquired using digital video analysis with the open source software Tracker.

Introduction

Oscillations and harmonic motion are important in physics and engineering. Oscillating water columns,[1] swinging pendulums,[2] masses attached to springs,[3-5] and other methods[6] are the most widely used examples. Collection and analysis of the experimental data are normally done using potentiometers,[2] force sensors,[3-5] webcams,[6] photocells,[7] and photosensors.[8]

Theories and models

A typical simple pendulum consists of a rigid light wire from which a massive object, like a sphere, is suspended at one end, as shown in Fig. 1. The forces acting on the object are: tension force $\boldsymbol{T}$ from the wire, the fluid drag force $\boldsymbol{F}_D$, gravitational force (weight) $\boldsymbol{F}_g$ acting on mass m, and the buoyancy force $\boldsymbol{B}$ as described by Archimedes' principle.

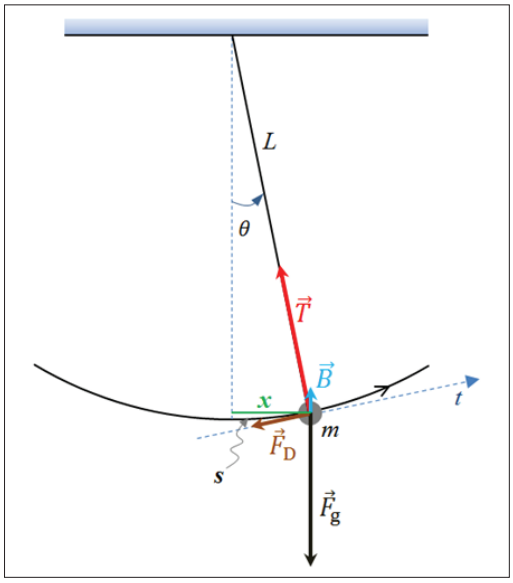

Fig. 1. Force diagram for a spherical pendulum in a fluid.

In the tangential direction t, components of the three forces acting on the sphere are: drag force $\boldsymbol{F}_D$, projection of the weight $\boldsymbol{F}_g$, and projection of the buoyancy force $\boldsymbol{B}$.

The magnitude of the weight and the buoyancy force are given by:

$$F_g = mg \tag{1}$$

$$B = \rho_f g V, \tag{2}$$

where g is the gravitational acceleration, ρ_f is the fluid density, and m is the mass of the sphere. The sphere volume V can be expressed in terms of its diameter d by:

$$V = \frac{1}{6}\pi d^3. \tag{3}$$

The differential equation of the oscillatory motion in the tangential direction t, considering viscous damping, is derived using Newton's second law of motion, $\Sigma \boldsymbol{F}_i = m\,\boldsymbol{a}$:

$$\boldsymbol{B}\sin\theta - F_g \sin\theta - F_D = m\,a, \tag{4}$$

where a represents the tangential acceleration. Equation (4) can be expressed in terms of the deflection angle θ as:

$$B\sin\theta - F_g\sin\theta - F_D = m\left(L\frac{d^2\theta}{dt^2}\right), \tag{5}$$

where L is the length of the wire.

Substituting Eqs. (1) and (2) into Eq. (5) and rearranging,

$$m\left(L\frac{d^2\theta}{dt^2}\right) = \left(\rho_f \tfrac{1}{6}\pi d^3 - m\right) g\sin\theta - F_D. \tag{6}$$

For small angular displacement, the deflection angle θ is very small and $\theta \cong \sin\theta = \frac{x}{L}$.

The sphere position, represented by s, is measured along the arc. Because $s = \theta L$, for small deflection angle, $s \cong x$ and $x \cong \theta L$. Thus, for small θ, the sphere oscillates almost in the horizontal direction, and Eq. (6), which describes the sphere's motion through the fluid, can be simplified to:

$$m\frac{d^2x}{dt^2} = \frac{g}{L}\left(\rho_f \tfrac{1}{6}\pi d^3 - m\right)x - F_D. \tag{7}$$

The sphere's oscillating motion depends on the drag force F_D. This force depends on the sphere's velocity. The Reynolds number (Re) can be used to relate viscosity of the fluid with the sphere's velocity:

$$Re = \frac{d\rho_f v}{\mu}, \tag{8}$$

where μ is the fluid viscosity and v is the relative speed between the fluid and the sphere.

Depending on the value of Reynolds number, different considerations can be taken into account to analyze the motion. These considerations are well explained in a publication by J. J. Mendoza-Arenas et al.,[4] where the use of different approaches to study the oscillating motion and calculate the

01 April 2026 11:37:20

fluid's viscosity led to the three different methods described in the appendix to this paper and which we used to determine viscosity, as follows:

- ***Method I –*** Stokes' law uses a term that is proportional to the sphere's speed v;
- ***Method II –*** Landau–Lifshitz model uses a much more complicated expression for F_D than Method I and should give better results in terms of calculating the fluid viscosity;
- ***Method III –*** Landau–Lifshitz model + corrective term uses a similar approach to Method II but takes into account the relative dimensions of the sphere and the liquid container.

The difference between the three models is the drag force term F_D in Eq. (7). The derivations of the three methods are presented in the appendix.[9]

Experimental setup

An experiment setup was used to measure the viscosity based on the above models. As shown in Fig. 2, a steel sphere with a (19.04 ± 0.01) mm diameter and having a mass of (28.343 ± 0.001) g is firmly attached to a rigid and lightweight wire of ≈ 2.8 m length. The wire has a mass of 1.201 g, 24 times less than the mass of the sphere m.

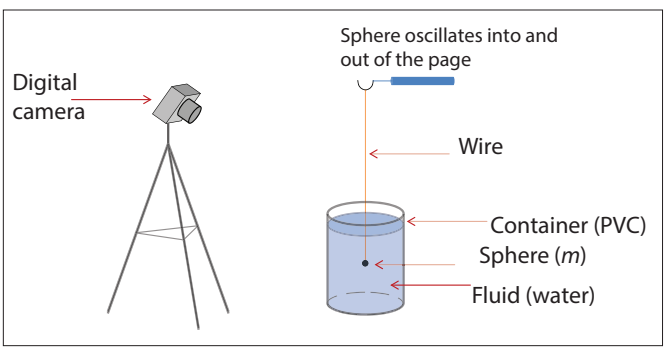

Fig. 2. Experimental setup (not all elements are to scale).

This pendulum is immersed inside a cylindrical container containing 650 mL of tap water. The container diameter is $D = (111.12 \pm 0.01)$ mm and height is (70.92±0.01) mm, at a temperature of 19.0 °C. We measured the mass of the volume of the water, thus obtaining the value of 998 kg/m^3 for its density. The pendulum is placed in horizontal oscillation and a digital camera (Canon PowerShot A550) acquires a time series of frames (about 720 frames), equivalent to 30 s of a digital movie, which corresponds to a nominal frame acquisition rate of 24 frames per second. It is important to set the camera so as to only capture the horizontal displacement in the x-direction. Using this digital video, it is possible to obtain the horizontal displacement x around its equilibrium position as a function of time.

The video analysis was done using Tracker video analyzer. Tracker is a free video analysis and modeling tool built on the Open Source Physics (OSP) Java framework. The "Autotracker" option allows collection of position coordinate

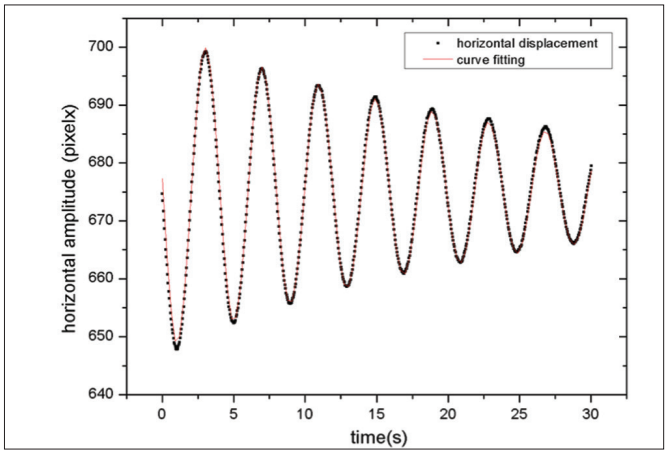

Fig. 3. Data from trial number 1 are shown, damped simple harmonic oscillation vs. time (dots). The red curve is theoretical fit given by Eq. (9). The adjusted *R*-squared obtained was 0.9985.

data (pixel-x) and the oscillation time. This way, we removed much of the tedium often involved in such an experiment. The data were analyzed/plotted using commercially available software[10] and fit to the damped harmonic oscillator curve given by

$$x = Ae^{-bt}\cos(\omega t - \phi), \tag{9}$$

so as to obtain the values of the parameters A (amplitude), ω (frequency), b (drag coefficient), and ϕ (phase shift); see the online appendix, Eq. (14).[9]

Using Eqs. (12), (18), and (20) from the appendix[9] along with the estimate for the drag coefficient, it is possible to calculate the viscosity of the water for the three models. The average viscosity and standard deviation were obtained from seven trials.

Data analyses

Figure 3 shows the experimental results of sphere oscillation (points) vs. time for the immersed sphere in tap water and the theoretical comparison (red curve) of Eq. (9). The maximum speed measured was 1.02×10^{-2} m · s^{-1} and the Re values were calculated to be less than 190, but greater than 1, so, according to considerations of Eq. (9) in the appendix[9], Stokes' law and Method I should not provide reliable estimates for viscosity.

The fitting coefficients A (amplitude), b (damping constant), ω (angular frequency), and ϕ (phase) are shown in Table I for seven trials. For all trials, the sphere was released from a distance less than its diameter to get very low amplitude oscillations. Viscosity is calculated using Methods I, II, and III and are shown in Table II. In the appendix[9] we show one example of how to determine viscosity from the fitting coefficients.

For distilled water at 19 °C the viscosity is 1.03×10^{-3} Pa.[11] As shown in Table II, viscosity calculated using Method I – Stokes' law has too high values, with a mean of 1.138×10^{-2} Pa·s. With Method II the mean value is 1.67×10^{-3} Pa·s, well above 1.03×10^{-3} Pa·s. Method III, 1.00×10^{-3} Pa·s was closest to published values, with a percentage error of 3%.

01 April 2026 11:37:20

Table I. Parameters from the curve fitting of experimental data.

Trial	A (pixel-x)	b (s^{-1})	ω (rad · s^{-1})	ϕ (rad)	Correlation coefficient R^2
1	27.52±0.06	0.0372±0.0002	1.5845±0.0002	4.641±0.002	0.9985
2	23.98±0.04	0.0351±0.0001	1.5850±0.0001	4.388±0.002	0.9991
3	24.87±0.03	0.0353±0.0001	1.5849±0.0001	4.641+0.001	0.9997
4	26.95±0.05	0.0353±0.0001	1.5861±0.0001	3.921±0.002	0.9991
5	23.90±0.10	0.0339±0.0004	1.5848±0.0004	1.925±0.005	0.9944
6	31.87±0.04	0.0366±0.0001	1.5848±0.0001	5.644±0.001	0.9995
7	42.51±0.05	0.0385±0.0001	1.5847±0.0001	3.871±0.001	0.9997

Table II. Viscosities obtained for the tap water using the three models.

Trial (movie)	Method I: *Stoke's law* Viscosity, μ (Pa·s) $\times 10^{-2}$	Method II: *Landau-Lifshitz* Viscosity, μ (Pa·s) $\times 10^{-3}$	Method III: *Landau–Lifshitz +corrective term* Viscosity, μ (Pa·s) $\times 10^{-3}$
1	1.185	1.79	1.08
2	1.109	1.59	0.95
3	1.116	1.61	0.96
4	1.115	1.60	0.96
5	1.071	1.49	0.89
6	1.156	1.71	1.03
7	1.216	1.88	1.13
Mean	**1.138**	**1.67**	**1.00**
Standard deviation	**5.0×10^{-4}**	1.3×10^{-4}	**8.0×10^{-4}**

The same method was used in seven trials for liquid ethanol 70% V/V, at 19.4 °C. The values obtained for the mean viscosity and standard deviation of the experimental data for the viscosities were:

Method I – **Stokes' law:** μ = (0.0178±0.0006) Pa·s,

Method II – **Landau-Lifshitz:** μ = (0.0041±0.0002) Pa·s

Method III – **Landau-Lifshitz** + corrective term: μ = (0.0025±0.0002) Pa·s

From Ref. 11, the viscosity for ethanol at 20 °C is: (0.00237) Pa·s, for ethanol 74.80%V/V and (0.00267) Pa·s, for ethanol 65.56% V/V.

We can see that the viscosity obtained experimentally (19.4 °C) was μ = (0.0025±0.0002) Pa·s, which is within the accepted range for the 70%V/V ethanol.

Conclusions

A conventional digital camera and computer software were used to study the damped harmonic motion of an underwater pendulum oscillating in tap water to measure fluid viscosity. The underwater pendulum is a useful and simple system to run at different levels of education and for applications in the classroom, allowing the study of laws about damped harmonic motion. The experiment is very easy to perform and requires inexpensive materials, besides a video camera, although a webcam or a smartphone can be used instead. The automated methods of data collection remove much of the tedium associated with video analysis and allow accurate measurements by students. Three models were used to estimate fluid viscosity. One estimate was obtained by Stokes' law, the other by Landau-Lifshitz model equations, and another was determined by Landau-Lifshitz with a corrective term. The Landau-Lifshitz with a corrective term model gives the best values for viscosity: 1.00 × 10^{-3} Pa·s for distilled water at 19 °C and 0.0025 Pa·s for ethanol 70%V/V at 19.4 °C.

We suggest applying the immersed pendulum method to other fluids, such as glycerin and motor oil, and up in the air as well, and analyze the movement by video analysis. These kinds of studies help students to understand the application of theoretical physics models in real applications, and the method presented here enables the study of a wide variety of sphere-fluid combinations for Reynolds numbers ranging from 0 to 190.

Acknowledgments

We're very grateful to English teachers Alexandrina Vila Franca and Fernanda Maria de Oliveira for helping us translate this paper.

References

1. J. E. Gilson and O. A. Boedtker, "A damped harmonic motion experiment for use in undergraduate general physics laboratories," *Am. J. Phys.* **37**, 1157–1158 (Nov. 1969).
2. M. F. McInerney, "Computer-aided experiments with the damped harmonic oscillator," *Am. J. Phys.* **53**, 991–996 (Oct. 1985).
3. P. Alexander and E. Indelicato, "A semi-empirical approach to a viscously damped oscillating sphere," *Eur. J. Phys.* **26**, 1–10 (Oct. 2004).
4. J. J. Mendoza-Arenas, E. L. D. Perico, and F. Fajardo, "Motion of a damped oscillating sphere as a function of the medium viscosity," *Eur. J. Phys.* **31**, 129–141 (Nov. 2009).
5. Sohaib Shamim, Wasif Zia, and Muhammad Sabieh Anwar, "Investigating viscous damping using a webcam," *Am. J. Phys.* **78**, 433–436 (April 2010).
6. R. D. Edge, "Viscosity," *Phys. Teach.* **20**, 47-48 (Jan. 1982).
7. M. Allen and E. J. Saxl, "The period of damped simple harmonic motion," *Am. J. Phys.* **40**, 942–944 (July 1972).
8. V. K. Gupta, G. Shanker, and N. K. Sharma, "Experiment on fluid drag and viscosity with an oscillating sphere," *Am. J. Phys.* **54**, 619–622 (July 1986).
9. Readers can download an appendix at *TPT Online* under the "Supplemental" tab, http://dx.doi.org/10.1119/1.5011833.
10. OriginPro 8.0, OriginLab Corp., Northampton, MA.
11. E. C. Bingham and R. F. Jackson, "Standard substances for the calibration of viscometers," *Bull. Bur. Stand.* **14**(1), 59–86 (April 1918).

01 April 2026 11:37:20